

Wind Tunnel Investigation of Spoiler Effectiveness on a low Aspect-Ratio Swept Wing with reflex Airfoils

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INTRODUCTION & AIM

During the development phase of a new radio-controlled aircraft at the Politecnico di Milano, characterized by a 40° sweep angle, a pronounced taper ratio, and a very low aspect ratio, it became necessary to evaluate the implementation of roll control using either conventional ailerons or spoilerons.

To support this design choice, a preliminary comparison was conducted using analytical methods (DATCOM), which are traditionally based on conventional airfoil data. However, a distinctive feature of this compact aircraft under development introduced the need to validate these preliminary estimates: the aircraft employs experimental reflex airfoils for which no prior data are available.

These factors led to the necessity of conducting wind tunnel tests in order to validate the predictions obtained from analytical methods for this class of airfoils, while simultaneously assessing roll performance. This work was also carried out to extend the conventional chordwise and spanwise ranges for spoileron placement.

The results were then evaluated through the introduction of performance indices, designed to provide the aircraft designer with sensitivity parameters related to geometry and spoiler positioning, ultimately supporting the optimization of spoiler sizing and placement on the wing during the development phase.

METHOD

The experimental campaign was conducted in a wind tunnel featuring a square test section with a side length of 2.4 m, operating at a wind velocity of 30 m/s, corresponding to an equivalent Reynolds number of 1e6.

The model was set at the centre of the test section on a 1,2 m high pylon, while the data acquisition system was installed on the wind tunnel wall at the base of the pylon.

The test model was machined from solid aluminium. The experimental campaign was carried out by installing spoilers manufactured from Plexiglas and secured using FDM 3D-printed supports at various locations along the upper surface of the right wing.

The spoilers were characterized by the following parameters: the ratio between the spoiler chord and the local wing chord, the ratio between the spoiler span and the wingspan, the geometric position along the wing, and the spoiler deflection angle.

A performance mapping of the spoiler was first conducted by maintaining a constant spoiler-to-wing chord ratio, of 0.15, defined during the preliminary design phase of the RC model, while varying the spoiler starting position both in the spanwise ($\frac{b_s}{b}$ from 0.4 to 0.6) and chordwise directions ($\frac{c_s}{c}$ from 0.4 to 0.6). Subsequently, at selected locations, the investigation was extended with a sensitivity analysis by varying either the spoiler-to-wing chord ratio, the spoiler span ratio or the deflection was carried out.

RESULTS & DISCUSSION

Figure 1 illustrates the discretization used for spoiler performance mapping. The orange region represents the domain covered by DATCOM data, highlighting the extension investigated in this study, with test points moved toward the leading edge.

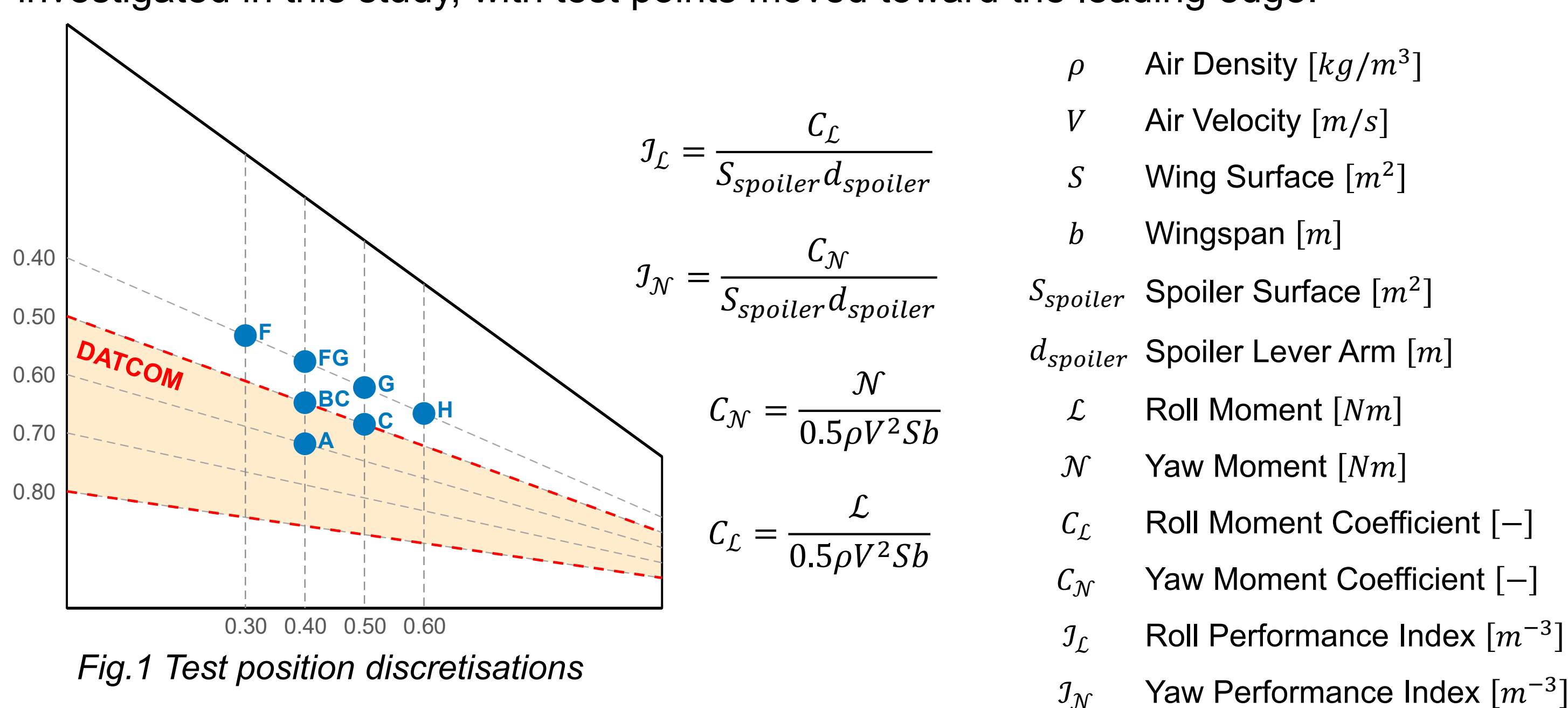


Fig.1 Test position discretisations

As shown in Fig. 2a, the preliminary campaign includes only three angles of attack (0°, 8°, 16°). Results are consistent with NACA/NASA findings, showing peak performance at intermediate angles of attack, followed by a decrease.

For spoileron design purposes, the effects of positioning and geometry may overlap, making it difficult to identify an optimal design strategy. For this reason, the merit indexes J_L and J_N were introduced to filter out the influence of the spoiler arm $d_{spoiler}$ and its surface area $S_{spoiler}$. This approach enables the identification of an optimal configuration independent of geometric scaling effects. As shown in Fig. 2a, doubling the chord (and thus the surface area) leads to a significant performance improvement. However, Fig. 2b shows that, when the indexes are introduced, a comparable performance increase, at angles of attack close to the maximum values suggested by NACA, can be achieved through a slight increase in the spoiler spanwise extension rather than in chord, while keeping the chord equal to the baseline configuration.

From a design standpoint, the results suggest that, for a given lever arm, increasing the spoileron span is more effective than increasing its chord, leading to reduced surface area and lower actuation force requirements.

Figure 3a and 3b present the mapping results at 8°, corresponding to the best-performing intermediate angle of attack. All test cases share the same spoileron configuration ($C_s/C = 0.15$, deflection 50°, $b_s/b = 0.3$). A clear performance peak is observed at position C, located beyond the conventional DATCOM applicability range.

Spoileron	Base	Span+	ΔArea	Chord+	ΔArea
Area [m ²]	0.0094	0.0126	+34%	0.0189	+100%

Tab.1 Total spoiler surface area variation

CONCLUSIONS

This preliminary study, to be further expanded, shows behaviour consistent with conventional airfoils in terms of aircraft roll and yaw response. The proposed merit index J_L and J_N also supports an alternative design strategy, identifying the most effective parameters for spoileron sizing optimization. Based on these results, improving roll or yaw performance is more effectively achieved by increasing the span rather than the chord, thereby limiting the increase in spoileron surface area and, consequently, the required actuator force.

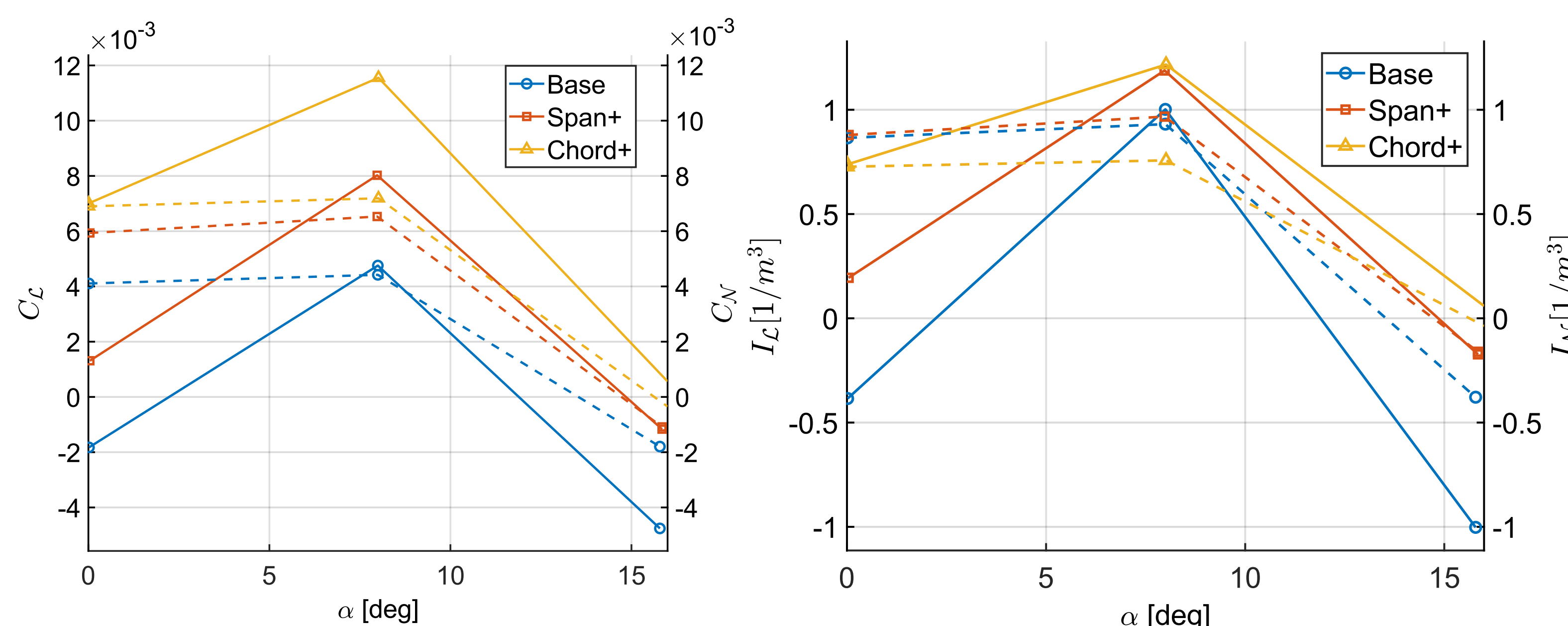


Fig.2a C_L and C_N evaluated in position H

Fig.2b J_L and J_N evaluated in position H

The presence of a maximum at mid-span is also consistent with the predictions provided by DATCOM. However, the observed maximum is in an extended region compared to the area previously tested.

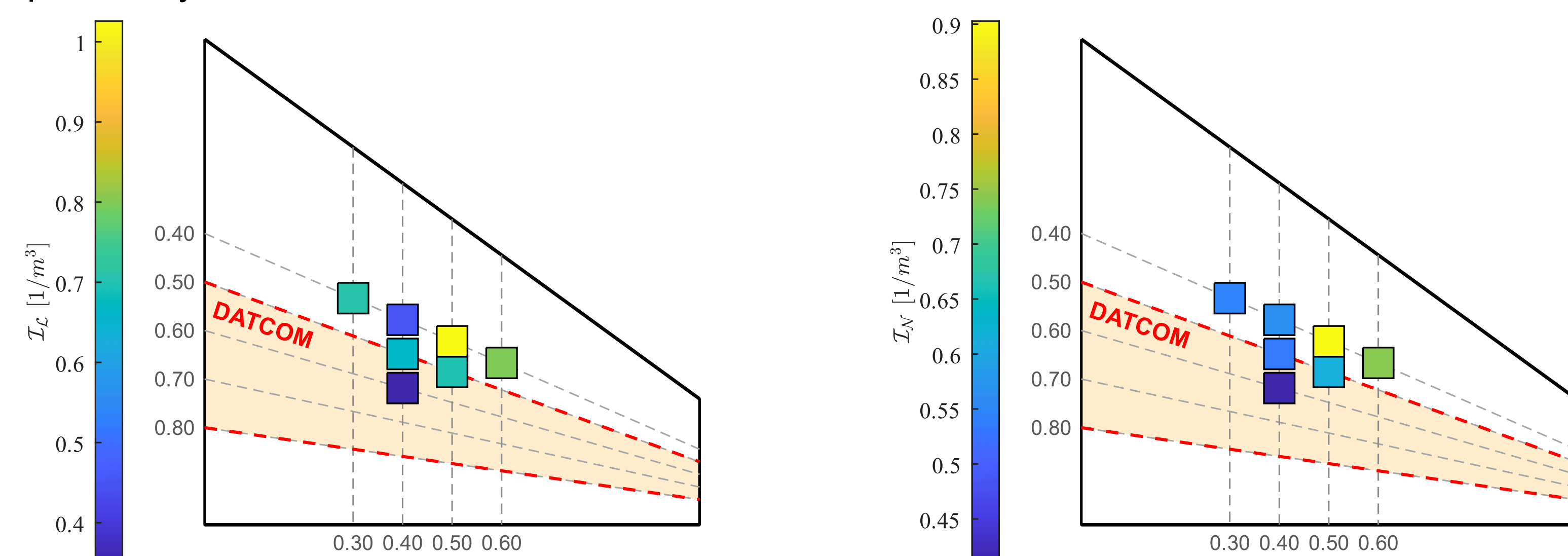


Fig.3a J_L mapped at $\alpha = 8^\circ$

Fig.3b J_N mapped at $\alpha = 8^\circ$

In terms of $C_{L_{\delta s}}$ and $C_{N_{\delta s}}$, an additional study was carried out at point A by varying the spoiler deflection in order to compare DATCOM predictions with the experimental results. The trend is found to be linear. For the rolling moment coefficient, discrepancies are present but remain limited, whereas the yawing moment coefficient is significantly mispredicted.

	$C_{L_{\delta s}}$ [1/rad]	Error [%]	$C_{N_{\delta s}}$ [1/rad]	Error [%]
DATCOM	0.0177	-	0.0043	-
WTT – 0°	0.0200	+ 13.00	0.0064	+ 48.83
WTT – 10°	0.0191	+ 7.91	0.0088	+104.65

Tab.2 Comparison WT results with DATCOM prediction

FUTURE WORK/ REFERENCES

Future developments of this work will include a finer discretization of the roll and yaw curves as functions of angle of attack, a more detailed analysis of the roll and yaw coefficients with respect to spoiler deflection, and an assessment of the impact on the overall aircraft lift.

«Data on spoiler type ailerons», John G. Lowry, NACA RM L 53124, 1953

«Investigation of effect of span, spanwise location, and chordwise location of spoiler on lateral control characteristics of a tapered wing», Jack Fischel and Vito Tamburello, NACA TN 1294, 1947

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Wind Tunnel Investigation of Spoileron Effectiveness on a Low-Aspect-Ratio Swept Wing with Reflex Airfoils [†]

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Abstract

Politecnico di Milano is undergoing the design of a highly swept, low-aspect-ratio radio-controlled aircraft with reflex airfoils. This model is necessary to expand the automated flight-testing activities conducted inside the university, adding to the flying models a more unconventional one to verify the flight-testing technique implemented. Plain-type spoilerons were investigated as primary roll control devices and compared with conventional aerodynamic predictions and wind-tunnel data. The experimental tests assessed performance across spanwise and chordwise positions, angles of attack, and spoileron geometric variations. A normalized control effectiveness parameter, accounting for moment coefficient, spoileron surface area, and moment arm, was introduced to compare configurations. Results show consistent peak performance at intermediate incidence and highlight distinct degradation patterns near stall. Spanwise variations primarily affect roll authority, while yaw response remains weakly sensitive. Geometric analysis indicates span increases are more efficient than chord increases for equivalent performance, reducing actuator loads and aerodynamic penalties.

Keywords: reflex-airfoil; wind-tunnel; spoileron; experimental; low aspect ratio; swept-wing

1. Introduction

During the development phase of a new radio-controlled aircraft characterized by a highly swept wing ($\Lambda \sim 45^\circ$), strongly tapered planform (~ 0.3), and limited aspect ratio ($\frac{b^2}{S} = 4$), a study on control surfaces was undertaken starting from traditional preliminary performance estimation methods, such as the DATCOM [1] method, followed by a comparison with experimental measurements obtained in a wind tunnel. The need for this experimental investigation arose from the necessity to assess the influence of a different family of airfoils on the analytical predictions provided by DATCOM. In fact, the experimental spoileron investigations reported in the literature have been conducted exclusively using conventional airfoils, whereas the present test article is characterized by the adoption of reflex airfoils.

This comparative analysis gains further significance considering that the airfoils adopted are reflex airfoils ($t/c \sim 10\%$) of innovative origin, derived from the practical field of flying-plank configurations [2]. Consequently, these airfoils must also be regarded as highly experimental.



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In particular, because the aft portion of the airfoil plays a decisive role in its self-stabilizing characteristics [3], its interaction with the spoiler is of particular interest and warrants investigation, as this region is the most strongly influenced by spoileron deflection, which induces flow separation [4].

The need to investigate not only the actual aerodynamic performance compared with the predicted one but also the interaction between these unconventional airfoils and spoileron-based control solutions motivated the execution of a dedicated wind-tunnel experimental campaign. Furthermore, an additional objective of this investigation was to improve the understanding of aircraft design tools for spoileron sizing, enabling the optimization of actuator requirements while maximizing aerodynamic performance.

The analyzed performances concern the capability of spoilers to generate rolling moment and the associated cross-coupling effects on yaw motion. Moreover, the study aimed to develop a performance map of the wing by varying the spoiler position both spanwise and chordwise, as well as modifying the geometric parameters of the spoileron itself, in order to evaluate how these variations could influence the resulting aerodynamic performance.

2. Testing Method

The model (represented in Figure 1) was positioned at the center of a square test section with a side length equal to 1.8 times the wingspan. The dimensions of the test section were selected to provide sufficient clearance between the wing tips and the tunnel walls, thereby minimizing wall-interference effects as suggested by literature [5]. The flow velocity was set at 30 m/s, corresponding to an equivalent Reynolds number of 930,000 [6].

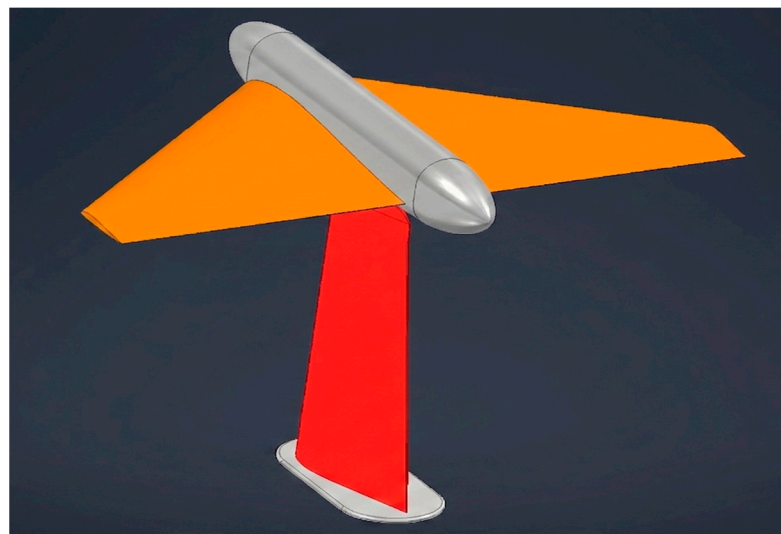


Figure 1. Experimental wind tunnel test setup.

The data acquisition system was installed on the wind tunnel wall at the base of the support pylon, while the aerodynamic loads were measured using an external balance. The support pylon was enclosed within an aerodynamic fairing/shield to prevent its aerodynamic contribution from being included in the force and moment measurements.

The test model, machined from solid aluminum, consisted of two semi-wings mounted on a fuselage that is not geometrically representative of the actual aircraft configuration. This fuselage was specifically designed to minimize aerodynamic interference during the tests and to allow the installation of an electric motor for additional experiments related to propeller slipstream.

The experimental campaign was carried out by installing plain spoilerons manufactured from Plexiglas and secured using FDM 3D-printed supports (as shown in Figure 2b) at various locations along the upper surface of the right wing (as shown in Figure 2a).

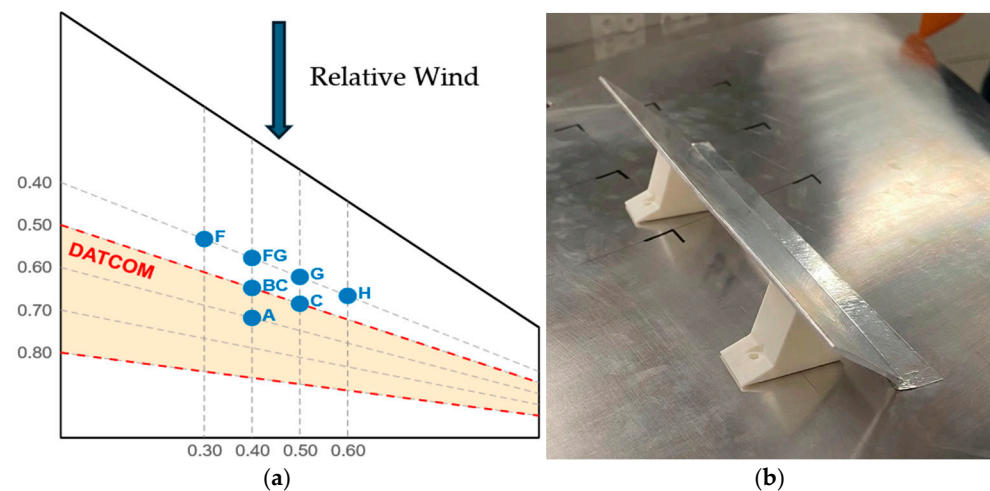


Figure 2. (a) Wing map of the tested points related to the DATCOM validity range; (b) Example of spoileron installation on the model.

The plain spoilerons were characterized by the following parameters: the ratio between the spoiler chord and the local wing chord $\frac{C_{\text{spoileron}}}{C}$, the ratio between the spoiler span and the wingspan $\frac{b_{\text{spoileron}}}{b}$, the geometric position along the wing chord or span, and the spoiler deflection angle $\delta_{\text{spoileron}}$.

A performance mapping of the plain-type spoiler was first conducted by maintaining constant spoiler-to-wing chord and span ratios, defined during the preliminary design phase of the RC model, while varying the spoiler starting position both in the spanwise (from 0.3 to 0.6 of the span) and chordwise directions (from 0.4 to 0.6), using the root-leading-edge corner of the spoileron as the reference point for its positioning along the wing.

It is important to note that some of the tested configurations lie outside the DATCOM reference validity range (0.5 to 0.8 wing-chord [7]). For this reason, these test points proved particularly useful for extending the results obtained through interpolation from the regression models provided by the analytical methods, thereby assessing their applicability and potentially extending the validity range of the DATCOM approach.

Subsequently, at selected locations, the investigation was extended by varying either the spoiler-to-wing chord ratio (named Chord+) or the span ratio (named Span+). Upon completion of the performance mapping over the wing, a sensitivity analysis with respect to the spoiler deflection angle was also performed at the preliminary design position.

3. Results

The experimental campaign allowed a detailed evaluation of spoileron performance across a range of spanwise and chordwise positions, spoiler geometries, and angles of attack. The analysis revealed several consistent trends regarding roll and yaw control authority.

As shown in Figure 3, the position of the spoiler along the wingspan was found to strongly influence roll effectiveness. At low angles of attack, spoilerons located closer to the wing tip generally produced larger rolling moments due to the greater moment arm relative to the aircraft's longitudinal axis. Interestingly, the outermost spoilerons did not always correspond to the maximum effectiveness; rather, a mid-span location, approximately

corresponding to point G in the experimental layout, provided the optimal compromise between roll authority and robustness against stall effects.

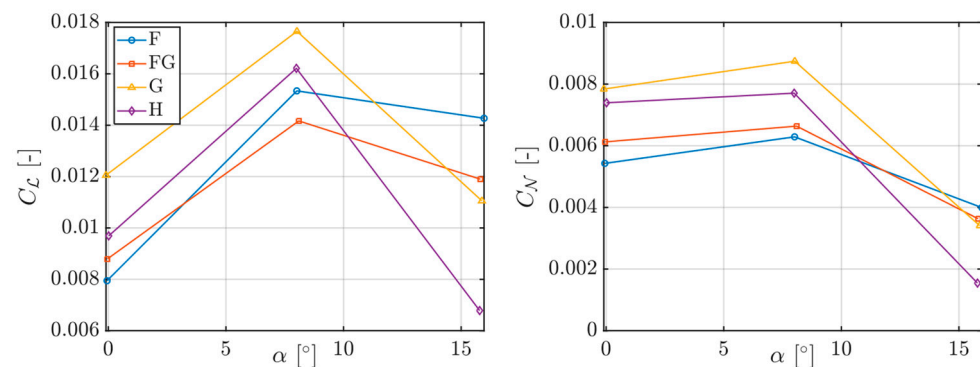


Figure 3. Roll and yaw performance while changing the spoileron spanwise position.

As the angle of attack increased, all spanwise configurations exhibited a peak in effectiveness at intermediate incidence, followed by a reduction in performance as the wing approached stall conditions, as stated also by Lowry in his work [8] (Figure 4).

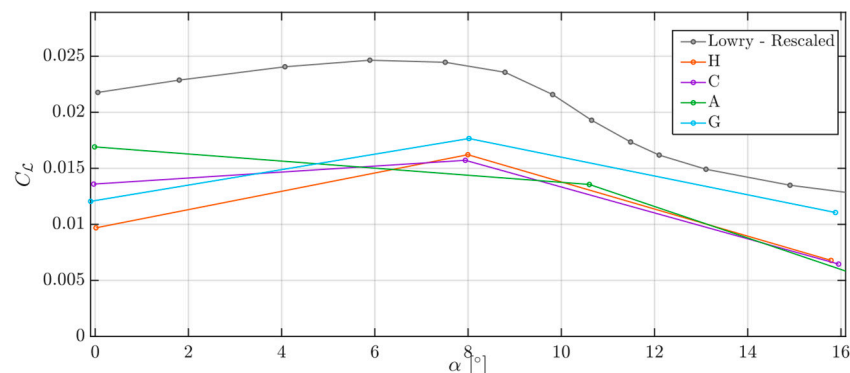


Figure 4. Rolling coefficient at different alpha compared with the trend highlighted by Lowry.

The outer spoilerons experienced an earlier degradation, likely due to the onset of flow separation in the tip regions, whereas more inboard spoilerons maintained partial effectiveness over a wider range of incidence. These findings underscore the importance of mid-span spoiler placement for achieving stable control in highly swept, low-aspect-ratio wings.

The effect of the spoiler chordwise position was therefore investigated for both roll and yaw control. This analysis was also motivated by the fact that, in DATCOM, the estimation of yawing effectiveness is treated in a more simplified manner than the corresponding roll analysis. Consequently, a more detailed assessment was performed to better characterize the resulting performance trends. Furthermore, it was hypothesized that flow development in the vicinity of stall conditions could significantly influence the effectiveness of spoilers located further aft along the chord, providing an additional motivation for investigating the impact of spoiler chordwise position.

At low angles of attack (Figure 5), the best performance is obtained with the spoilers located further aft along the chord, closer to the trailing edge. At high angles of attack, however, the best performance is achieved with spoilers positioned further forward, closer to the leading edge. In the intermediate angle-of-attack range, a convergence of performance among configurations is expected. Unfortunately, due to the experimental campaign being conducted with different angles of attack for the A position, a complete comparison at intermediate incidence is unavailable due to a difference in the alpha of testing.

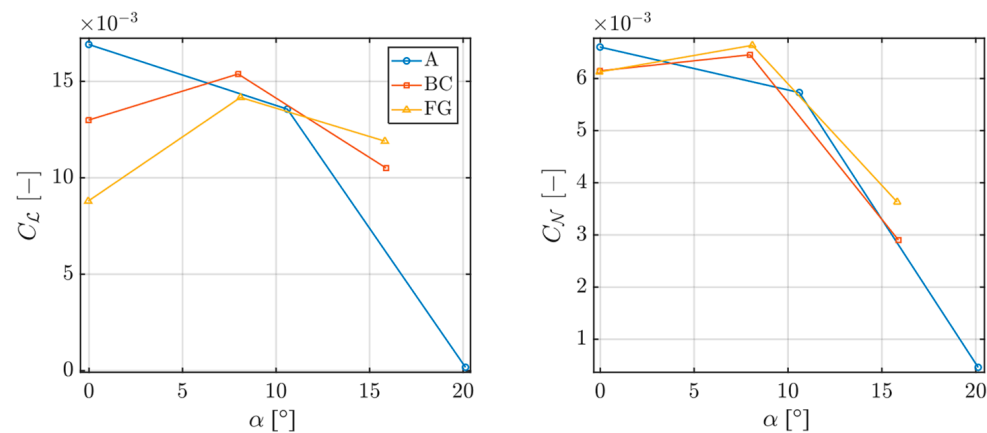


Figure 5. Roll and yaw performance while changing the spoileron chordwise position.

This behavior may be interpreted as a greater resistance, for forward-mounted spoilerons, to the upstream propagation of the separation bubble originating from the trailing edge at the wingtip region during the stall.

The geometric characteristics of the spoiler itself further modulated performance. Increasing either the chord or span resulted in enhanced roll authority, as expected. The influence of spoiler geometry was investigated at the A and H test locations through a dedicated sensitivity analysis. Two geometric modifications were considered: the spoiler chord ratio was increased from 0.15 to 0.30 (doubling the spoileron area), and the spoiler span was increased by approximately 34% while maintaining the original chord ratio.

As expected, both modifications led to an increase in roll-control effectiveness, as shown in Figure 6. In particular, the increase in spoiler chord produced a larger enhancement in roll performance than that obtained through the corresponding span increase, having a larger total surface.

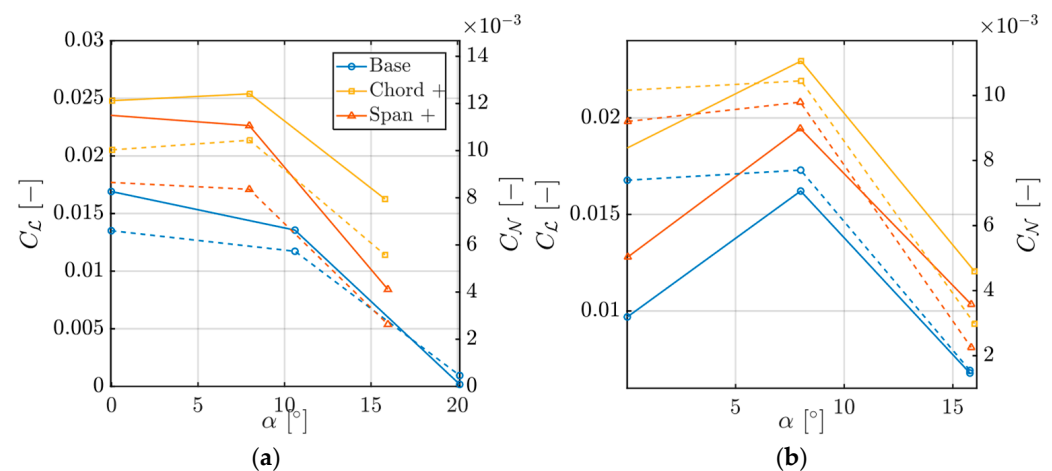


Figure 6. Roll (solid line) and yaw (dashed line) while changing a geometric parameter of the spoileron: (a) spoileron in position A, (b) spoileron in position H.

This observation suggests that the relationship between spoiler geometry and aerodynamic effectiveness cannot be assessed solely on the basis of the absolute increase in rolling moment. The comparison between the two geometric modification strategies therefore provides the basis for the discussion presented in the following section, where a normalized effectiveness metric is introduced to support design-oriented evaluations.

At positions A and H, a study was also conducted on roll performance as a function of surface deflection at angles of 15°, 30°, and 50° (Figure 7). For angles of attack of 0° and

10°, the response is essentially linear. However, at 20°, beyond the aircraft's stall angle, roll performance is almost entirely absent, as the spoilers have lost all authority.

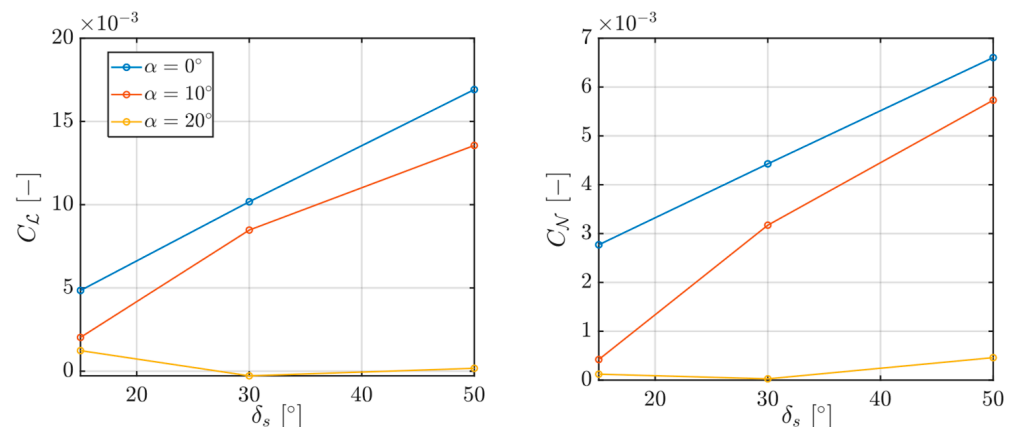


Figure 7. Rolling and yawing coefficients, changing the spoileron deflection.

When comparing these experimental results with the roll performance previously estimated using DATCOM, listed in Table 1, discrepancies are observed, although they are not excessive. At this stage of the study, it is difficult to determine whether these differences are due to the presence of the reflex airfoil or the wing planform. Further investigations will be carried out to clarify this point.

Table 1. Error DATCOM estimation and WTT measurements.

Source	$C_{L_{\delta s}}$ Error [%]	$C_{N_{\delta s}}$ Error [%]
DATCOM	-	-
WTT 0°	+13.00	+48.83
WTT 10°	+7.91	+104.65

4. Discussion

During data post-processing, two additional spoileron performance indices, $\mathcal{I}_L$ and $\mathcal{I}_N$, were introduced (Equations (3) and (4)). These indices were conceived as design-oriented metrics to compare different spoileron locations based on the effectiveness generated by a unit spoiler area $S_{spoiler}$ acting through a unit moment arm $d_{spoiler}$ with respect to the center of gravity.

$$C_L = \frac{\mathcal{L}}{0.5\rho V^2 S b} \quad (1)$$

$$C_N = \frac{\mathcal{N}}{0.5\rho V^2 S b} \quad (2)$$

$$\mathcal{I}_L = \frac{C_L}{S_{spoiler} d_{spoiler}} \quad (3)$$

$$\mathcal{I}_N = \frac{C_N}{S_{spoiler} d_{spoiler}} \quad (4)$$

They therefore provide a useful tool for identifying the regions of the wing that offer the highest control efficiency, supporting both aerodynamic interpretation and preliminary spoileron design.

The results (Figure 8) show that, for both positions A and H, at intermediate angles of attack (corresponding to the previously identified peak performance), this parameter

exhibits overlap between tests with a doubled chord and tests with only a 34% increase in span.

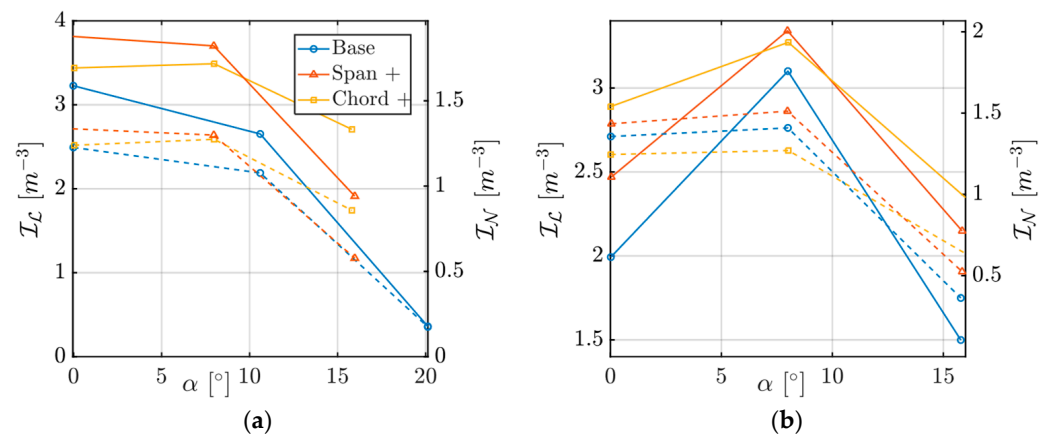


Figure 8. Roll (solid line) and yaw (dashed line) while changing a geometric parameter of the spoileron and introducing the I index: (a) Position A; (b) Position H.

This indicates that a smaller increase in spoileron span can achieve similar performance to doubling the chord but with reduced aerodynamic penalty (drag) as listed in Table 2.

Table 2. Geometrical variations from the reference model.

Spoileron	Chord+	$\Delta\mathcal{I}_L$ [%]	Span+	$\Delta\mathcal{I}_L$ [%]
Position A	+100% $S_{spoiler}$	+22.86	+34% $S_{spoiler}$	+63.37
Position H	+100% $S_{spoiler}$	+21.44	+34% $S_{spoiler}$	+30.03

Therefore, designers can infer that, to achieve the same control effectiveness, it is preferable to operate on the spoileron span rather than the chord, thereby reducing actuator forces and overall system weight.

In the following images (Figures 9–11), it can be observed that the optimal spoiler placement remains within the central region of the test points. This represents the ideal compromise between performance at zero angle of attack and near-stall conditions, ensuring a delayed loss of control authority. Furthermore, it is evident that due to the stall phenomenon, the optimal point gradually shifts inward and toward a more forward position along the wing as the flow approaches separation conditions.

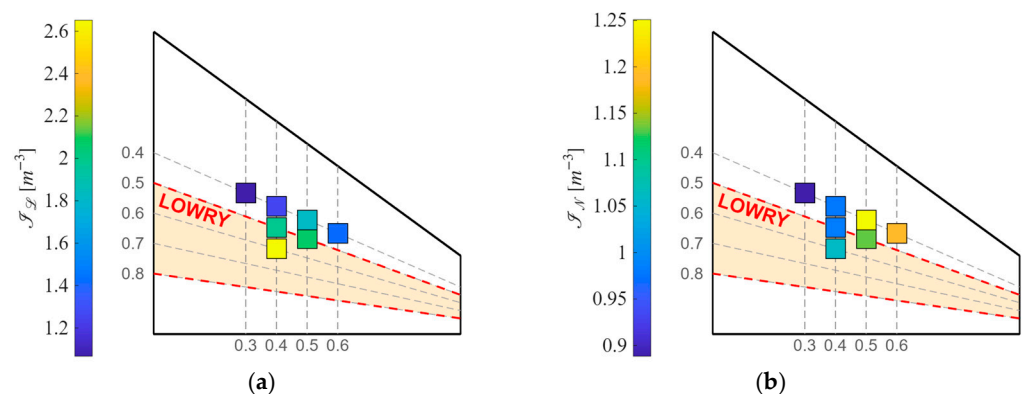


Figure 9. $\alpha = 0^\circ$ Performance index I: (a) Roll performance; (b) Yaw performance.

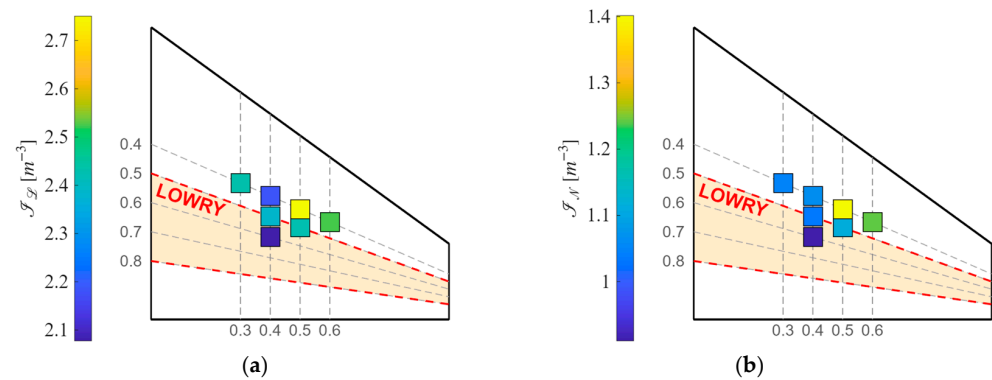


Figure 10. $\alpha = 8^\circ$ Performance index I: (a) Roll performance; (b) Yaw performance.

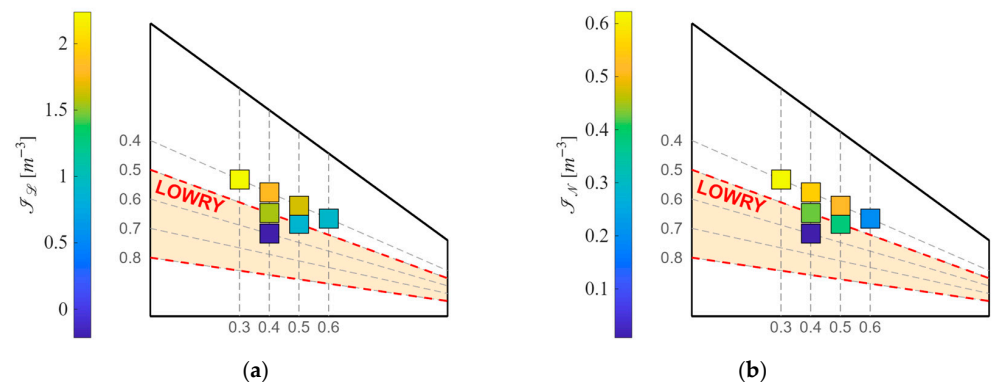


Figure 11. $\alpha = 16^\circ$ Performance index I: (a) Roll performance; (b) Yaw performance.

5. Conclusions

The present experimental campaign investigated the use of spoilerons as primary roll-control devices for a highly swept, low-aspect-ratio wing equipped with reflex airfoils. The wind-tunnel measurements provided an opportunity to evaluate both the applicability of conventional preliminary prediction methods and the influence of spoiler location and geometry on the resulting control authority.

The results showed that the overall behavior of the spoilerons remains consistent with trends commonly reported for conventional wing configurations. In particular, roll effectiveness was found to increase with angle of attack up to an intermediate-incidence condition, beyond which a degradation occurs as the wing approaches stall. While spoilerons positioned closer to the wing tip generate larger rolling moments at low incidence because of their greater moment arm, they also experience a more rapid loss of effectiveness as the angle of attack increases. This behavior is likely associated with the earlier onset of flow separation in the outer wing sections.

The experimental mapping of spoiler position highlighted the existence of a preferred installation region located approximately at the wing mid-span, as similarly shown in DATCOM and Tamburello's report. The configuration identified as point G consistently provided the most favorable compromise between low-incidence effectiveness and robustness at higher angles of attack. This result is in agreement with trends reported in the available literature on spoiler-based roll-control systems.

The comparison between different spoiler geometries further showed that both chord and span increases lead to improved control authority. However, when the achieved performance is normalized through the proposed effectiveness index, span increases appear to provide a more efficient solution than chord increases. From a design perspective, this suggests that equivalent aerodynamic performance may be obtained with reduced actuation requirements and lower system weight by favoring spoiler span rather than chord.

Finally, although the motivation for the study originated from the use of reflex airfoils, no substantial deviations from the behavior expected for conventional airfoils were observed within the investigated test envelope. The present work should therefore be regarded as a first experimental assessment of spoileron performance on this class of unconventional configuration. Future investigations will focus on a finer discretization of spoiler geometry and position, as well as on a more detailed analysis of spoiler deflection effects.

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Conflicts of Interest: The authors declare no conflict of interest.

Nomenclature

ρ	Air Density [kg/m ³]
V	Air Velocity [m/s]
S	Wing Surface [m ²]
b	Wing Span [m]
$S_{spoiler}$	Spoiler Surface [m ²]
$d_{spoiler}$	Spoiler Lever Arm [m]
$\mathcal{L}$	Roll Moment [Nm]
$\mathcal{N}$	Yaw Moment [Nm]
$C_{\mathcal{L}}$	Roll Moment Coefficient [–]
$C_{\mathcal{N}}$	Yaw Moment Coefficient [–]
$\mathcal{I}_{\mathcal{L}}$	Roll Performance Index [m ^{−3}]
$\mathcal{I}_{\mathcal{N}}$	Yaw Performance Index [m ^{−3}]

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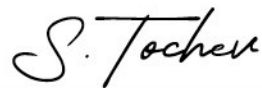
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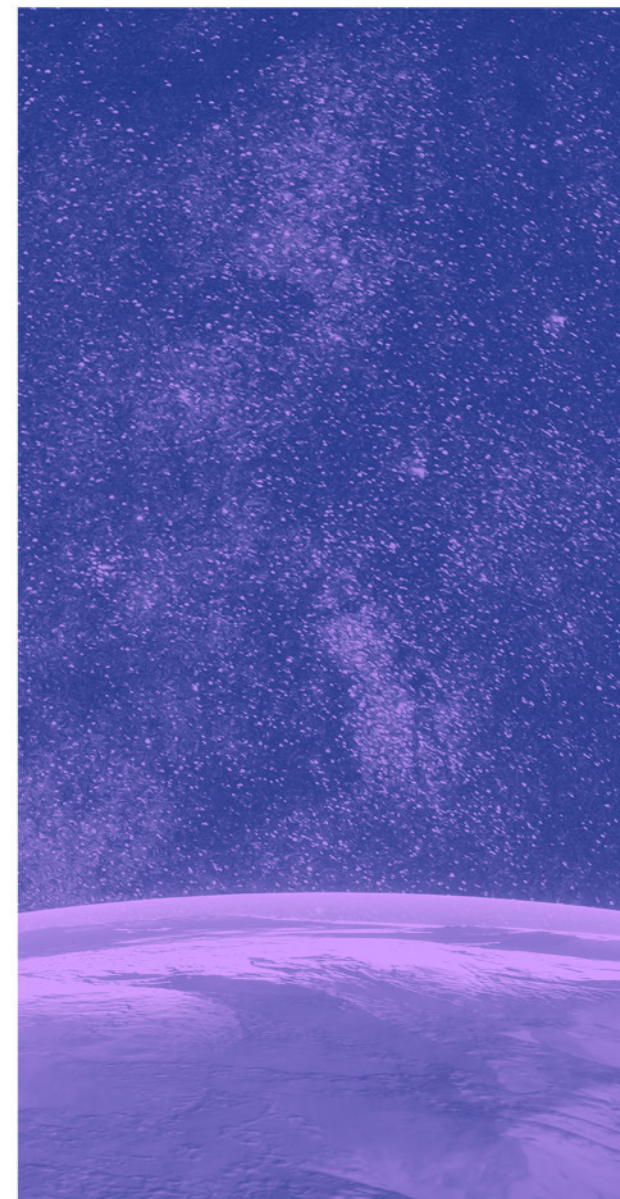
Certificate of Participation

This certificate is hereby presented to the attendee
named below in recognition of their active participation.

Filippo Coacci



Stefan Tochev
Chief Executive Officer



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